

MODULE 38

SQL and Query Performance

Master SQL fundamentals and advanced optimization techniques to improve query performance, reduce response time, and build efficient database solutions in enterprise environments.







MODULE 38 OVERVIEW

Master SQL fundamentals and advanced optimization techniques to improve query performance, reduce response time, and build efficient database solutions.

Every application depends on the database. This module covers writing correct SQL, combining tables with joins, using indexes and execution plans, and tuning real PostgreSQL queries -- backed by a hands-on evidence-based lab.

DURATION & LAB



1 Hour Theory

SQL review, all four JOIN types, indexes, execution plans, and query tuning strategies for PostgreSQL.



2 Hours Lab

PostgreSQL SQL and CRM Query Performance: real DBMS_XPLAN plans, justified indexes, sargable rewrites, and paging.



Scenario

As a Database Performance Engineer for an e-commerce platform, analyze slow queries and optimize database objects.

MODULE ARC



Master SQL & Joins

Review SELECT/WHERE/ORDER BY and combine tables with all four JOIN types.



Indexes & Plans

Understand clustered/non-clustered indexes and read EXPLAIN / EXPLAIN ANALYZE plans.



Tune & Monitor

Apply tuning strategies, avoid full table scans, and monitor SQL performance in production.

KEY TAKEAWAY Total time: 3 hours (1 hour theory + 2 hours hands-on lab). By the end, you will write efficient SQL, optimize performance, and keep PostgreSQL databases fast, reliable, and scalable.

WHY QUERY OPTIMIZATION MATTERS (1 OF 2)

Every application depends on the database. Inefficient SQL queries can slow down systems, increase costs, and impact user experience.

Optimized queries ensure your database performs at its best -- today and as your data grows. A single `SELECT * FROM orders WHERE status = 'Shipped'` can be fast or crippling depending on how it's written.

THE COST OF POOR QUERIES



Slow Response Times

Users experience delays and timeouts, leading to frustration and lost productivity.



Higher Infrastructure Costs

Inefficient queries consume more CPU, memory, and I/O, requiring more hardware.



Poor Scalability

As data grows, poorly written queries degrade performance and limit your ability to scale.



Business Impact

Slow reports, delayed decisions, and missed SLAs can directly affect business outcomes.



Harder to Maintain

Complex, unoptimized queries are difficult to maintain and increase the risk of errors.

KEY TAKEAWAY Slow queries don't just feel sluggish -- they chain directly into higher infrastructure costs, worse scalability, real business impact, and harder maintenance.

WHY QUERY OPTIMIZATION MATTERS (2 OF 2)

The benefits of query optimization are the direct mirror image of the costs on the previous slide -- faster, cheaper, more scalable, more reliable.

THE BENEFITS OF QUERY OPTIMIZATION



Faster Performance

Return results quickly and improve overall application speed.



Lower Costs

Use resources efficiently and reduce infrastructure and operational costs.



Better Scalability

Handle more data and users without sacrificing performance.



Reliability & Stability

Consistent performance helps meet SLAs and improves system reliability.



Better User Experience

Happy users, faster workflows, and stronger business impact.

KEY TAKEAWAY Optimizing your SQL queries is not just a technical task -- it's a critical factor in delivering fast, reliable, and cost-effective database solutions that drive business success.

MODULE LEARNING OBJECTIVES

By the end of this module, you will be able to write efficient SQL, optimize performance, and ensure your PostgreSQL databases are fast, reliable, and scalable.

- **1. Understand SQL Fundamentals** — review core SQL concepts including SELECT statements, filtering, sorting, and basic query structure.
- **2. Work with Joins** — use INNER, LEFT, RIGHT, and FULL OUTER JOINS to combine data from multiple tables effectively.
- **3. Understand Indexes** — explain the role of indexes and differentiate between clustered and non-clustered indexes.
- **4. Analyze Query Execution** — read and interpret query execution plans to understand how queries are processed.
- **5. Identify Performance Issues** — identify slow queries and understand common causes such as missing indexes and full table scans.
- **6. Apply Optimization Techniques** — use query tuning strategies to improve query performance and reduce resource consumption.
- **7. Avoid Full Table Scans** — design queries and indexes that help the database engine retrieve data efficiently.
- **8. Design Efficient Queries** — write clear, efficient, and maintainable SQL that delivers accurate results with optimal performance.
- **9. Follow Database Best Practices** — apply enterprise-level best practices for SQL design, indexing, and performance tuning in production.
- **10. Monitor SQL Performance** — use tools and techniques to monitor, measure, and maintain database performance in production.

KEY TAKEAWAY Module Outcome: you will be able to write efficient SQL queries, optimize performance, and ensure your PostgreSQL databases are fast, reliable, and scalable.

SQL REVIEW

SQL (Structured Query Language) is the standard language used to communicate with relational databases -- store, retrieve, update, and manage data efficiently.

SQL CATEGORIES

Category	Examples
DDL — Data Definition	CREATE, ALTER, DROP, TRUNCATE
DML — Data Manipulation	INSERT, UPDATE, DELETE
DQL — Data Query	SELECT
DCL — Data Control	GRANT, REVOKE
TCL — Transaction Control	COMMIT, ROLLBACK, SAVEPOINT

BASIC SQL SYNTAX

```
SELECT column1, column2, ...  
FROM table_name  
WHERE condition  
GROUP BY column_name  
HAVING condition  
ORDER BY column_name ASC|DESC  
LIMIT number;
```

EXAMPLE: SIMPLE SELECT QUERY

```
SELECT customer_id, first_name, last_name, email  
FROM customers  
WHERE status = 'Active'  
ORDER BY last_name ASC  
LIMIT 10;
```

SELECT

FROM

WHERE

JOIN

GROUP BY

HAVING

ORDER BY

DISTINCT

LIMIT

KEY TAKEAWAY SQL is the foundation of working with relational databases. In this module, we build on these basics and focus on writing high-performance PostgreSQL SQL.

SELECT STATEMENTS

The SELECT statement retrieves data from one or more tables. It is the most frequently used SQL statement.

BASIC SYNTAX

```
SELECT column1, column2, ...  
FROM table_name;
```

- **SELECT** — specifies the columns to retrieve.
- **column1, column2, ...** — one or more column names, or * for all columns.
- **FROM** — specifies the table (or view) to retrieve data from.

COMMON VARIATIONS

Variation	Example
Remove duplicates	SELECT DISTINCT status FROM orders;
Limit rows	SELECT * FROM orders LIMIT 10;
Filter + sort	SELECT * FROM orders WHERE status = 'Shipped' ORDER BY order_date DESC;

EXAMPLES

Description	SQL Query
Select all columns	SELECT * FROM customers;
Select specific columns	SELECT first_name, email FROM customers;
Select with column alias	SELECT first_name AS name, email AS contact FROM customers;

KEY TAKEAWAYS

- SELECT is used to retrieve data from the database.
- You can select all columns (*) or specific columns.
- Use aliases (AS) to make result sets easier to read.
- Combine with WHERE, ORDER BY, and LIMIT to get the exact data you need.

KEY TAKEAWAY Best Practice: always select only the columns you need instead of using * to improve performance and reduce network traffic.

FILTERING DATA WITH WHERE

The *WHERE* clause filters rows based on a specified condition. Only rows where the condition evaluates to *TRUE* are returned.

COMMON COMPARISON OPERATORS

Operator	Example
=	WHERE age = 30
<> or !=	WHERE status <> 'Closed'
> / <	WHERE amount > 1000
>= / <=	WHERE salary >= 50000
BETWEEN	WHERE age BETWEEN 18 AND 30
IN	WHERE status IN ('New','Open')
LIKE	WHERE name LIKE 'A%'
IS NULL / IS NOT NULL	WHERE email IS NULL

EXAMPLES

Description	SQL Query
Filter by equality	WHERE city = 'Toronto'
Filter by range	WHERE order_date >= '2024-01-01' AND order_date <= '2024-03-31'
Multiple conditions	WHERE category = 'Electronics' AND price > 500
Filter using IN	WHERE status IN ('Active','Pending')
Filter NULL values	WHERE last_login IS NULL

KEY TAKEAWAYS

- WHERE filters rows based on conditions.
- Combine operators and logical keywords (AND, OR, NOT) for complex filters.
- Filtering data early improves performance by reducing the result set size.

KEY TAKEAWAY Best Practice: use indexes on columns used in the WHERE clause to improve query performance.

SORTING DATA WITH ORDER BY

The *ORDER BY* clause sorts the result set in ascending (*ASC*) or descending (*DESC*) order based on one or more columns.

BASIC SYNTAX

```
SELECT column1, column2, ...  
FROM table_name  
ORDER BY column1 [ASC|DESC], column2 [ASC|DESC];
```

- **ASC** — ascending order is the default.
- **DESC** — descending order, most-recent/highest first.
- **Multiple columns** — sorted left to right in priority order.
- **NULL handling** — whether NULLs sort first or last is database dependent.

BEST PRACTICE

Always order by indexed columns when possible, especially on large datasets, to avoid an expensive in-memory sort.

EXAMPLES

Description	SQL Query
Sort one column (ASC)	ORDER BY first_name ASC
Sort one column (DESC)	ORDER BY first_name DESC
Sort multiple columns	ORDER BY city ASC, order_date ASC
Mixed ASC and DESC	ORDER BY city ASC, order_date DESC

KEY TAKEAWAYS

- ORDER BY is used to sort the result set.
- ASC sorts ascending (default), DESC sorts descending.
- You can sort by multiple columns to achieve more specific ordering.
- Sorting large result sets can be resource intensive — use indexes and filters.

KEY TAKEAWAY ORDER BY sorts the result set by one or more columns — sort by indexed columns when possible on large datasets to avoid an expensive in-memory sort.



INNER JOIN

An INNER JOIN returns rows that have matching values in both tables based on the join condition. Non-matching rows are excluded.

SYNTAX

```
SELECT a.column1, b.column2, ...
FROM table_a a
INNER JOIN table_b b
    ON a.common_column = b.common_column;
```

KEY POINTS

- Returns only matching rows from both tables.
- Eliminates rows with NULLs in the join column from either table.
- The most commonly used join type.
- Efficient when proper indexes exist on the join columns.

USE CASES

Order + customer details

Valid relationships only

EXAMPLE — CUSTOMERS INNER JOIN ORDERS

customer_id	first_name	order_id	amount
101	Aman	2001	125.00
101	Aman	2003	320.50
103	Priya	2002	250.75

EXPLANATION

- Matches customers with their orders on customer_id.
- Customers 102 and 104 have no orders, so they are excluded.
- Order 2004 references customer_id 105, which does not exist, so it is excluded too.

KEY TAKEAWAY Best Practice: ensure indexes exist on join columns (e.g., customer_id) in both tables to keep INNER JOIN performance efficient.



LEFT JOIN

A LEFT JOIN returns all rows from the left table and the matching rows from the right table. Unmatched right-table columns return NULL.

SYNTAX

```
SELECT a.column1, b.column2, ...  
FROM table_a a  
LEFT JOIN table_b b  
  ON a.common_column = b.common_column;
```

KEY POINTS

- Preserves all rows from the left table (table_a).
- Fills unmatched right-table columns with NULL.
- Useful for finding related data or identifying missing matches.
- Not always symmetrical — the order of tables matters.

USE CASES

All customers + orders (if any)

Customers with no orders

EXAMPLE — CUSTOMERS LEFT JOIN ORDERS

customer_id	first_name	order_id	amount
101	Aman	2001	125.00
101	Aman	2003	320.50
102	Neha	NULL	NULL
104	John	NULL	NULL

EXPLANATION

- Customer 101 has two orders → two rows returned.
- Customers 102 and 104 have no orders → order columns show NULL, but the customer row is kept.

KEY TAKEAWAY Best Practice: ensure an index exists on the right table's join column (e.g., orders.customer_id) to keep LEFT JOIN performance efficient.

RIGHT JOIN

A *RIGHT JOIN* returns all rows from the right table and the matching rows from the left table. Unmatched left-table columns return *NULL*.

SYNTAX

```
SELECT a.column1, b.column2, ...
FROM table_a a
RIGHT JOIN table_b b
  ON a.common_column = b.common_column;
```

KEY POINTS

- Preserves all rows from the right table (table_b).
- Fills unmatched left-table columns with NULL.
- Useful when the right table is the primary focus of the query.
- Not always symmetrical — the order of tables matters.

USE CASES

All orders, even unmatched

Right table as primary dataset

EXAMPLE — CUSTOMERS RIGHT JOIN ORDERS

customer_id	first_name	order_id	amount
101	Aman	2001	125.00
103	Priya	2002	250.75
101	Aman	2003	320.50
NULL	NULL	2004	150.25

EXPLANATION

- Order 2004 belongs to customer_id 105, which does not exist in customers.
- Since it has no match in the left table, the customer columns are NULL — but the order row is still kept because it exists in the right table.

KEY TAKEAWAY Best Practice: ensure indexes exist on join columns in both tables (customers.customer_id, orders.customer_id) to keep RIGHT JOIN performance efficient.

FULL OUTER JOIN

A *FULL OUTER JOIN* returns all rows from both tables — matching rows, plus non-matching rows from each side with *NULLs* filling the gaps.

SYNTAX

```
SELECT a.column1, b.column2, ...
FROM table_a a
FULL OUTER JOIN table_b b
  ON a.common_column = b.common_column;
```

KEY POINTS

- Returns all rows from both tables.
- Includes matching rows AND non-matching rows from both sides.
- Non-matching columns are filled with NULL.
- Useful for data comparison, reconciliation, and completeness checks.

BEST PRACTICE

- Use COALESCE() to handle NULLs in display or sorting.
- Be cautious with FULL OUTER JOIN on large tables — it can be expensive.
- Ensure proper indexing on join columns to improve performance.

EXAMPLE — CUSTOMERS FULL OUTER JOIN ORDERS

customer_id	first_name	order_id	amount
101	Aman	2001	125.00
102	Neha	NULL	NULL
103	Priya	2002	250.75
NULL	NULL	2003	320.50
NULL	NULL	2004	150.25

LEGEND

Matched (both)

Left only

Right only

KEY TAKEAWAY FULL OUTER JOIN = INNER JOIN matches + LEFT-only rows + RIGHT-only rows, all in one result set — reserved for reconciliation and completeness checks.

TRY IT YOURSELF — EXERCISE 1: ACCESS PATTERNS

Reinforces: naming how the CRM will actually query customers and accounts before we learn indexes to support those queries.

STEPS (IN notes/lab38-perf.md)

Step	What To Do
1 — List Patterns	Write at least three real CRM access patterns: lookup by customer_id (CUS-1001), filter by status, filter by created_at range, and accounts by customer_id.
2 — Name the Hot Path	Mark the single hottest path: single-customer lookup by customer_id — the query every CRM screen calls first.
3 — Flag the Anti-Pattern	Write one line warning against SELECT * without a WHERE clause on a huge table in application code.
4 — Save Your Notes	Save the pattern list plus the named hot path to notes/lab38-perf.md.

Hot path vs. anti-pattern

```
-- Hot path: single-customer lookup
SELECT customer_id, full_name, status FROM customer WHERE customer_id = 'CUS-1001';
-- Anti-pattern: avoid in app code on a huge table
SELECT * FROM customer;
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 1 before continuing: exercises/exercise-01-access-patterns.md (workspace: examples/module-38-exercises).

UNDERSTANDING INDEXES

Indexes are special data structures that improve the speed of data retrieval operations on a database table.

An index works like the index in a book — it helps the database find data quickly without scanning the entire table.

WHAT IS AN INDEX?

- A data structure (e.g., B-Tree, Hash) that stores a small portion of the table data in a sorted way.
- Contains a copy of the indexed column value and a pointer to the actual row.
- The database uses it to locate rows quickly without scanning the whole table.

WHY INDEXES IMPROVE PERFORMANCE

- **Reduces full table scans** — database locates rows via the index instead of reading every row.
- **Speeds up filters and sorts** — helps WHERE, JOIN, ORDER BY, GROUP BY, DISTINCT.
- **Efficient data access** — less disk I/O and CPU usage.

TYPES OF INDEXES

Type	Used For
B-Tree	Most common; equality and range queries (>, <, BETWEEN, ORDER BY).
Hash	Equality (=) lookups only — not used for range queries.
Composite	Created on multiple columns, e.g. (last_name, first_name).
Unique	Ensures all values in the indexed column are unique.

TRADE-OFFS & CONSIDERATIONS

- Indexes improve reads but slow down writes (INSERT, UPDATE, DELETE).
- Indexes consume additional storage space.
- Too many indexes can degrade performance and increase maintenance overhead.
- Create indexes based on query patterns, not on every column.

KEY TAKEAWAY Indexes make queries faster by allowing the database to find data quickly — essential for large tables. Use them wisely to balance read performance and write overhead.

TRY IT YOURSELF — EXERCISE 2: INDEX TRADEOFFS

Reinforces: the Trade-Offs & Considerations you just saw on Understanding Indexes, applied to the real CRM write path.

STEPS (DOCUMENT THE TRADEOFF)

Step	What To Do
1 — Name the Benefit	Write the read-side win: faster status filters and faster account-by-customer joins.
2 — Name the Cost	Write the write-side cost: slower INSERT/UPDATE for the Amina/Ravi seed rows at scale, plus more disk used per index.
3 — Plan Cleanup	Note that the lab may drop experimental indexes — plan to document a before/after for each one you keep or remove.
4 — State the Rule	Write the rule you will follow all lab: add an index only when EXPLAIN shows it is actually needed.

The rule to write down

```
-- Add an index only when EXPLAIN shows a Seq Scan that a real query pattern actually needs fixed.
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 2 before continuing: [exercises/exercise-02-index-tradeoffs.md](#).

CLUSTERED VS NON-CLUSTERED INDEXES

Both index types improve query performance, but they store and organize data differently.

KEY DIFFERENCES

Feature	Clustered Index	Non-Clustered Index
Data storage	Data rows stored at the leaf level.	Pointers (row locators) stored at the leaf level.
Physical order	Determines the physical order of data.	Does not affect physical order.
Number per table	Only one.	Multiple allowed.
Performance	Faster for range queries and sorting.	Faster for searches, filters, and joins.
Space overhead	Lower.	Higher.

WHEN TO USE A CLUSTERED INDEX

- Column is unique and frequently used in range queries.
- Table is frequently read in sorted order.
- Primary key columns (automatically clustered in many databases).

WHEN TO USE A NON-CLUSTERED INDEX

- Columns used in WHERE, JOIN, GROUP BY, ORDER BY.
- Columns with high selectivity (return fewer rows).
- When you need multiple indexes on one table.

KEY TAKEAWAY Clustered Index = data is stored in index order (only one per table). Non-Clustered Index = index points to data (many allowed). Choose the clustered index column carefully — usually the primary key.

TRY IT YOURSELF — EXERCISE 3: FILL SQL/INDEX TODOS (STARTER)

Reinforces: CREATE INDEX syntax plus sargable filtering and paging — a fill-in-the-blank starter, done on paper, not run against Postgres.

STARTER FILE — notes/lab38-todos.sql (FILL EVERY BLANK)

```
-- baseline (avoid)
SELECT * FROM customer WHERE ____ (full_name) = 'amina khan';

-- optimized lookup
SELECT customer_id, full_name, status FROM customer WHERE customer_id = ____;

-- supporting index ideas
CREATE INDEX ____ ON customer (status);
CREATE INDEX ____ ON account (customer_id);

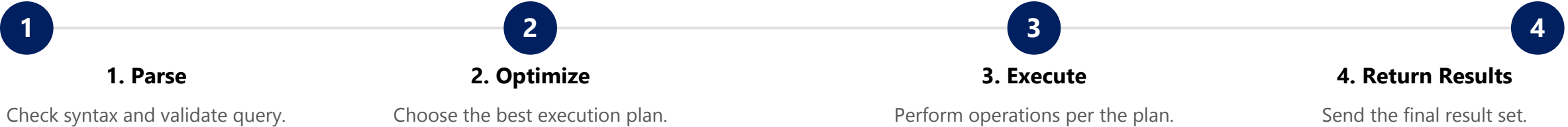
-- paging sketch
SELECT customer_id, full_name FROM customer ORDER BY customer_id LIMIT ____ OFFSET ____;
```

Blank	Suggested Fill
Baseline function / lookup value	lower — this is the non-sargable pattern to recognize, not repeat.
customer_id filter value	'CUS-1001' — Amina Khan's fixture.
Two index names	idx_customer_status, idx_account_customer.
LIMIT / OFFSET	20 / 0 — plus a // TODO comment: prefer keyset pagination (WHERE customer_id > :last) for deep pages.

KEY TAKEAWAY TRY IT NOW — Complete Exercise 3 (STARTER) before continuing: exercises/exercise-03-fill-sql-index-todos.md. This is a starter file, not a finished solution — replace every ____ and // TODO, and do not run it against Postgres yet; Lab 38 will measure it for real.

QUERY EXECUTION PLANS (1 OF 2)

A query execution plan shows how the database engine will execute a SQL query to return results — the roadmap for every tuning decision.



WHY EXECUTION PLANS MATTER

- Understand how the database retrieves data.
- Identify performance bottlenecks.
- Verify index usage.
- Compare different query versions.
- Make informed tuning decisions.

COMMON OPERATORS IN EXECUTION PLANS

Operator	Meaning
Seq Scan	Scans the entire table row by row; used when no index helps.
Index Scan / Seek	Uses an index to find matching rows — faster than a seq scan.
Nested Loop Join	For each outer row, finds matching inner rows.
Hash Join	Builds a hash table on one input and probes with the other.
Sort	Sorts rows based on specified columns.
Aggregate	Performs functions like COUNT, SUM, AVG.

KEY TAKEAWAY A query execution plan reveals exactly how the database retrieves your data — parse, optimize, execute, return. Reading it is the first step to fixing a slow query.

QUERY EXECUTION PLANS (2 OF 2)

Reading a real EXPLAIN ANALYZE plan turns a black box into a concrete, actionable tuning target.

EXAMPLE QUERY

```
SELECT e.name, d.dept_name FROM employees e JOIN departments d
ON e.dept_id = d.dept_id WHERE e.salary > 50000 ORDER BY e.name;
```

SAMPLE EXECUTION PLAN — EXPLAIN ANALYZE (POSTGRESQL)

```
Sort (cost=25.50..25.53 rows=12 width=64)
  Sort Key: e.name
  -> Hash Join (cost=0.42..25.32 rows=12 width=64)
    Hash Cond: (e.dept_id = d.dept_id)
    -> Seq Scan on employees e (cost=0.00..18.20 rows=20)
      Filter: (salary > 50000)
    -> Hash (cost=0.26..0.26 rows=26 width=36)
      -> Seq Scan on departments d (cost=0.00..0.26 rows=26)
```

READING THE PLAN — KEY COLUMNS

Column	Description
cost	Estimated cost to execute the operation (startup..total).
rows / width	Estimated number of rows returned and estimated row size in bytes.
actual time / loops	(With ANALYZE) real time taken and how many times the operation ran.

KEY TAKEAWAY Seq Scan applies the salary filter, Hash Join combines employees and departments, Sort orders by name — index WHERE/JOIN/ORDER BY columns and keep statistics current with ANALYZE.

TRY IT YOURSELF — EXERCISE 4: EXPLAIN CHECKLIST

Reinforces: reading the EXPLAIN ANALYZE plan you just saw, built into a reusable PostgreSQL-native checklist for Lab 38.

STEPS (BUILD THE CHECKLIST)

Step	What To Do
1 — Write the Command	Write the exact command you'll run in Lab 38: EXPLAIN (ANALYZE, BUFFERS) <sql>;
2 — Name What to Look For	List Seq Scan vs. Index Scan, the estimated rows, and the buffers line in your checklist.
3 — Name a Success Signal	Write: an Index Scan on customer_id for the Amina lookup is a good sign.
4 — Note the PostgreSQL Command	Write that ANALYZE customer; refreshes statistics in PostgreSQL — this is not DBMS_STATS.

The exact command

```
EXPLAIN (ANALYZE, BUFFERS) SELECT customer_id, full_name, status FROM customer WHERE customer_id = 'CUS-1001';
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 4 before continuing: [exercises/exercise-04-explain-checklist.md](#) (workspace: [examples/module-38-exercises](#)).

IDENTIFYING SLOW QUERIES (1 OF 2)

Slow queries degrade application performance and user experience. Identifying them early helps you optimize databases and ensure responsive systems.

WHAT IS A SLOW QUERY?

- Takes longer than expected to execute and consumes more database resources (CPU, I/O, memory).
- Impact: increased response time, higher load, timeouts, and poor user experience.

COMMON CAUSES

- Missing or ineffective indexes.
- Full table scans.
- Complex joins and subqueries.
- Unoptimized WHERE conditions.
- Large result sets.
- Outdated statistics.

WHERE TO LOOK

- **Database Logs** — enable the slow query log (e.g., `log_min_duration_statement` in PostgreSQL).
- **Query Monitoring Tools** — pgAdmin, DataGrip, DBeaver, or APM tools.
- **Application Logs** — identify queries taking too long in app logs.
- **Database Views** — system views for long-running or expensive queries.

PROCESS TO IDENTIFY SLOW QUERIES

- 1 **Detect**
- 2 **Capture**
- 3 **Analyze**
- 4 **Prioritize**
- 5 **Verify**

KEY TAKEAWAY A slow query takes longer than expected and consumes excess CPU, I/O, and memory — identify it early through logs, monitoring tools, and a repeatable detect-capture-analyze-verify process.

IDENTIFYING SLOW QUERIES (2 OF 2)

PostgreSQL ships built-in views and commands to find and analyze slow queries; combine them with disciplined best practices.

USEFUL POSTGRESQL VIEWS & COMMANDS

View / Command	What It Does
<code>pg_stat_statements</code>	Find top queries by total time, mean time, calls.
<code>EXPLAIN / EXPLAIN ANALYZE</code>	Show a query's execution plan and cost.
<code>pg_stat_activity</code>	View currently running queries and their duration.
<code>log_min_duration_statement</code>	Log queries that exceed a specified time (ms).
<code>auto_explain extension</code>	Automatically log slow queries with execution plans.

Example: Finding Slow Queries

```
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;
-- Top 5 slow queries by total time
SELECT query, calls, total_time, mean_time, rows
FROM pg_stat_statements
ORDER BY total_time DESC LIMIT 5;
```

WHAT TO CHECK IN THE EXECUTION PLAN

- Seq Scan instead of an Index Scan.
- High-cost operations (Sort, Hash, Materialize).
- Large gap between estimated vs. actual row count.
- Nested Loop on large datasets.
- Expensive filters executed after joins.

BEST PRACTICES

- Always use indexes on filter, join, and sort columns.
- Avoid SELECT *; fetch only required columns.
- Keep statistics updated (ANALYZE regularly).
- Use LIMIT for large result sets when applicable.
- Monitor and review slow queries continuously.

KEY TAKEAWAY Identify slow queries early with `pg_stat_statements` and `EXPLAIN ANALYZE`, analyze their root cause, and optimize for better performance.

QUERY TUNING STRATEGIES (1 OF 2)

Query tuning improves the performance of SQL queries to reduce response time and resource usage. Good queries are correct, efficient, scalable, and maintainable.

Faster Response Time

Lower Resource Usage

Higher Throughput

Better Scalability

Cost Efficiency



1. Use Indexes Effectively

Create indexes on WHERE/JOIN/ORDER BY/GROUP BY columns. Avoid over-indexing; use composite indexes for multi-column filters.



2. Write Efficient Queries

Select only required columns. Avoid functions on indexed columns. Prefer EXISTS over IN for large subqueries. Use LIMIT/OFFSET carefully.



3. Optimize Joins

Ensure join columns are indexed on both tables. Join smaller tables first. Prefer INNER JOIN when possible; avoid Cartesian products.

EXAMPLE — INEFFICIENT VS. EFFICIENT JOIN

```
-- Inefficient: SELECT * FROM orders o, customers c WHERE o.customer_id = c.id;
-- Efficient:  SELECT o.order_id, c.name FROM orders o
--             INNER JOIN customers c ON o.customer_id = c.id;
```

KEY TAKEAWAY Good query performance is critical for scalable, reliable applications — start by indexing effectively, writing efficient queries, and optimizing joins.

QUERY TUNING STRATEGIES (2 OF 2)

Four more strategies — reducing data processed, fresh statistics, execution plan analysis, and general best practices — complete the tuning toolkit.



4. Reduce Data Processed

Filter early with strong WHERE clauses. Partition large tables. Archive old data. Avoid unnecessary DISTINCT / ORDER BY on large result sets.



5. Update Statistics Regularly

The optimizer relies on statistics to choose the best plan. Outdated statistics lead to poor plans — run ANALYZE (PostgreSQL) regularly.



6. Analyze Execution Plans

Use EXPLAIN / EXPLAIN ANALYZE. Look for full scans, missing indexes, and expensive sorts/hashtables. Fix root causes, not symptoms.



7. Other Best Practices

Use prepared statements and bind variables. Avoid functions on indexed columns. Normalize data. Monitor and tune continuously.

EXAMPLE TUNING WORKFLOW

1

1. Identify Slow Query

2

2. Analyze Plan

3

3. Apply Strategies

4

4. Test & Validate

5

5. Monitor Continuously

KEY TAKEAWAY Use a combination of indexes, efficient SQL, and regular maintenance — always measure, analyze, and validate improvements. Small changes can lead to big performance gains.

AVOIDING FULL TABLE SCANS (1 OF 2)

A full table scan (sequential scan) reads every row in a table to find the required data. It is expensive and does not scale.

WHAT IS A FULL TABLE SCAN?

- The database reads every row sequentially.
- Used when no usable index exists.
- Happens with missing/unused indexes, functions on columns, or non-sargable predicates.
- Expensive and slow for large tables.

WHY AVOID FULL TABLE SCANS?

Slower Performance

High Resource Usage

Poor Scalability

Higher Cost

Bad User Experience

GOAL

Design queries and indexes so the database finds only the rows it needs — never the whole table.

COMMON REASONS FOR FULL TABLE SCANS

Reason	Explanation
Missing Index	No index on columns used in WHERE, JOIN, ORDER BY, or GROUP BY.
Functions on Columns	Wrapping a column in a function (e.g., UPPER(name)) prevents index usage.
Leading Wildcard	LIKE '%abc' cannot use a standard index.
Data Type Mismatch	Comparing different data types causes index invalidation.
Outdated Statistics	Optimizer may choose a full scan if statistics are stale.

KEY TAKEAWAY A full table scan reads every row and does not scale — it happens when no usable index exists, functions wrap indexed columns, or statistics are stale.

AVOIDING FULL TABLE SCANS (2 OF 2)

Sargable queries — ones the optimizer can match directly to an index — are the key to avoiding full table scans.

HOW TO AVOID FULL TABLE SCANS

- Create indexes on columns used in filtering, joining, sorting, and grouping.
- Write sargable queries (search ARGument-ABLE).
- Avoid functions on indexed columns in the WHERE clause.
- Use leading values in LIKE ('abc%' instead of '%abc').
- Use appropriate, consistent data types and comparisons.
- Keep statistics up to date (ANALYZE).

EXAMPLE — MAKING A QUERY SARGABLE

```
-- Non-sargable (causes full scan)
SELECT * FROM employees WHERE UPPER(name) = 'AMIT';
-- Sargable (uses index)
SELECT * FROM employees WHERE name = 'Amit';
```

HOW TO DETECT FULL TABLE SCANS

- **Execution Plan** — look for 'Seq Scan' in the plan.
- **Slow Query Logs** — identify queries with high duration and rows examined.
- **Monitoring Tools** — pg_stat_statements, EXPLAIN ANALYZE, APM tools.
- **High Rows Examined** — compare rows examined vs. rows returned.

BEST PRACTICES

- Design indexes based on real query patterns.
- Review slow queries and execution plans regularly.
- Avoid SELECT *; select only required columns.
- Partition large tables when appropriate.
- Monitor and tune queries continuously.

KEY TAKEAWAY Full table scans are one of the biggest performance killers. Sargable queries, proper indexing, fresh statistics, and regular monitoring keep the database reading only what it needs.

TRY IT YOURSELF — EXERCISE 5: SARGABILITY

Reinforces: classifying real predicates as sargable or not for PostgreSQL, using the UPPER(name) pattern you just saw.

REFERENCE TABLE — CLASSIFY EACH PREDICATE

Predicate	Sargable?
customer_id = 'CUS-1001'	Yes
status = 'ACTIVE'	Yes (with index)
LOWER(full_name) = 'amina khan'	Usually no on a plain index
created_at >= TIMESTAMP '2026-01-01'	Yes (range)
date_trunc('day', created_at) = ...	Often weaker than a range

STEPS

- **1 — Copy the table** — reproduce the reference table above into your notes.
- **2 — Rewrite one** — rewrite a non-sargable name search into something index-friendlier (store lowercased column, or use ILIKE carefully).
- **3 — State the range rule** — prefer created_at >= d AND created_at < d+1 over wrapping the column in a function.

KEY TAKEAWAY TRY IT NOW — Complete Exercise 5 before continuing: [exercises/exercise-05-sargability.md](#). If old materials say TRUNC(created_at), map it to this PostgreSQL range/date_trunc contrast.

EFFICIENT QUERY DESIGN (1 OF 2)

Efficient query design returns correct results using the least amount of time, resources, and I/O. Well-designed queries are readable, maintainable, and optimized.

Faster Performance

Lower Resource Usage

Better Scalability

Improved Reliability

Cost Efficiency



1. Select Only What You Need

Avoid SELECT *. Select only required columns to reduce data transfer and processing.



2. Use Proper Filtering

Use WHERE to filter early. Avoid functions on indexed columns.



3. Optimize Joins

Join only necessary tables. Ensure join columns are indexed. Use INNER JOIN unless an outer join is required.



4. Avoid Unnecessary Sorting

Use ORDER BY only when needed. Ensure sorting columns are indexed. Avoid sorting large result sets.

EXAMPLE — RETURN LESS DATA, FASTER PERFORMANCE

```
-- Inefficient: SELECT * FROM employees;  
-- Efficient:   SELECT emp_id, name, salary FROM employees;
```

KEY TAKEAWAY Efficient query design is about doing less work and returning results faster — select only what you need, filter early, optimize joins, and avoid unnecessary sorting.

EFFICIENT QUERY DESIGN (2 OF 2)

Four more techniques — smart aggregation, appropriate data types, avoiding subquery pitfalls, and leveraging indexes — complete a disciplined design workflow.



5. Use Aggregates Wisely

Filter rows before aggregation. Aggregate only necessary data. Avoid DISTINCT if GROUP BY is enough.



6. Use Appropriate Data Types

Use the smallest data type that fits. Avoid large types on indexed columns. Consistent types prevent implicit conversions.



7. Avoid Subquery Pitfalls

Use JOINS instead of correlated subqueries where possible. Ensure subqueries return only necessary rows.



8. Leverage Indexes

Create indexes on WHERE/JOIN/ORDER BY/GROUP BY columns. Avoid functions on indexed columns. Don't over-index.

EFFICIENT QUERY DESIGN WORKFLOW

1

1. Requirements

2

2. Filter Early

3

3. Join Smartly

4

4. Select Wisely

5

5. Check & Tune

6

6. Test & Monitor

KEY TAKEAWAY Efficient query design is not just about writing correct SQL — it's about writing smart SQL that does less work and returns results faster. Follow the workflow: filter, join, select, then test and tune.

DATABASE PERFORMANCE BEST PRACTICES (1 OF 2)

High-performing databases result from good design, efficient queries, proper indexing, regular maintenance, and continuous monitoring.

Better Performance

Efficient Resource Usage

Scalability

Reliability & Availability

Cost Efficiency



1. Design for Performance

Use appropriate data types, normalize data, choose the right keys, and consider partitioning for large tables.



2. Write Efficient Queries

Select only required columns, filter early with WHERE, use EXISTS over IN, use LIMIT/OFFSET carefully.



3. Use Indexes Wisely

Create indexes on WHERE/JOIN/ORDER BY columns; avoid over-indexing; rebuild or reorganize indexes regularly.



4. Avoid Expensive Operations

Avoid functions on indexed columns, implicit conversions, unnecessary sorting/DISTINCT, and Cartesian products.



5. Keep Statistics Up to Date

Update table and index statistics regularly — outdated statistics lead to poor execution plans.



6. Manage Resources

Monitor CPU/memory/disk/I/O usage, configure database parameters appropriately, use connection pooling.

KEY TAKEAWAY High-performing databases start with good design, efficient queries, proper indexing, and up-to-date statistics — small, consistent practices compound into big performance wins.

DATABASE PERFORMANCE BEST PRACTICES (2 OF 2)

The remaining six best practices — monitoring, transactions, maintenance, security, testing, and capacity planning — round out a production-grade performance program.



7. Monitor Continuously

Monitor slow queries and wait events; track performance metrics; set up alerts for anomalies; use dashboards.



8. Optimize Transactions

Keep transactions short; commit as early as possible; avoid long-running transactions.



9. Implement Proper Maintenance

Rebuild/reorganize indexes; clean up unused objects; archive old data; check and fix fragmentation.



10. Secure Your Database

Use strong passwords and roles; follow least privilege; encrypt data at rest and in transit; audit access.



11. Test and Tune Regularly

Test queries with realistic data volumes; review execution plans periodically; validate changes in lower environments.



12. Plan for Growth

Monitor growth trends; plan capacity for storage/memory/connections; use partitioning and archiving.

PERFORMANCE CHECKLIST

Slow queries reviewed

Indexes optimized

Statistics updated

Resources monitored

Maintenance performed

Security enforced

Growth planned

KEY TAKEAWAY Good database performance results from smart design, efficient queries, proper indexing, regular maintenance, and continuous monitoring — applied consistently to deliver fast, reliable, scalable solutions.

PRODUCTION SQL MONITORING (1 OF 2)

Production SQL monitoring helps you continuously track database health, detect issues early, and ensure consistent performance for your applications.

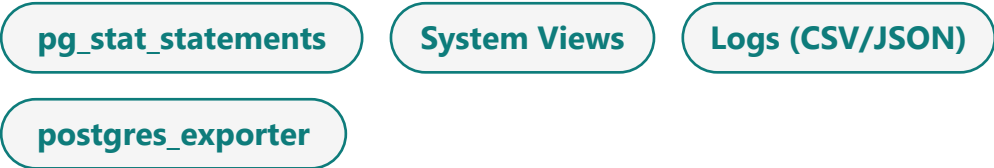
WHY MONITOR IN PRODUCTION?

- **Ensure Performance** — identify slow queries and bottlenecks before users are impacted.
- **Improve Availability** — detect locks, blocking, and resource contention early.
- **Reduce Costs** — optimize resource usage and avoid over-provisioning.
- **Capacity Planning** — understand trends to plan for growth and workload changes.

PRODUCTION MONITORING ARCHITECTURE



COLLECTION METHODS



KEY METRICS TO MONITOR

Metric	What It Tells You
Query Response Time (avg, p95, p99)	Overall query performance experienced by users.
CPU Utilization / Memory Usage	CPU pressure and buffer-cache / memory pressure on the database.
Lock Waits / Blocking / Deadlocks	Contention between concurrent queries and transactions.

KEY TAKEAWAY Monitor query response time, CPU/memory, and lock waits — the goal is to detect issues before users are impacted, not after.

PRODUCTION SQL MONITORING (2 OF 2)

Essential tools, alerting thresholds, and dashboards turn raw metrics into an early-warning system for database health.

ESSENTIAL TOOLS & SOURCES

- **pg_stat_statements** — tracks execution stats for all normalized queries.
- **System Views** — pg_stat_activity, pg_locks, pg_stat_io, pg_stat_database.
- **Logs** — log_min_duration_statement captures slow queries.
- **Monitoring Tools** — pgAdmin, DataGrip, DBeaver, Prometheus + Grafana, Datadog.

BEST PRACTICES

- Enable pg_stat_statements in all environments.
- Set and review slow query thresholds regularly.
- Correlate database metrics with application performance.
- Automate alerting and incident response.

COMMON ALERTS TO CONFIGURE

```
-- Slow queries: mean_time > 1000 ms
-- High CPU: cpu_usage > 80% for 5m
-- Lock waits: wait_event_type = 'Lock'
-- Deadlocks: deadlocks_total increases
-- Replication lag above threshold
```

DASHBOARDS TO INCLUDE

Top 10 Slowest Queries

Database Load

Active Connections

Lock Waits

Replication Status

Errors & Deadlocks

KEY TAKEAWAY Monitoring in production is not optional — it's essential for reliability, performance, and user satisfaction. Measure. Alert. Analyze. Optimize.

TRY IT YOURSELF — EXERCISE 6: LAB 38 READINESS (CAPSTONE)

Reinforces: every concept from this module, pulled into one final self-check before you open the graded Lab 38 guide.

STEPS (PRE-LAB SELF-CHECK)

Step	What To Do
1 — Write a Hypothesis	Write: customer_id equality uses an index; LOWER(full_name) may seq scan.
2 — Preview the Evidence	Note where it will live: future EXPLAIN snippets get pasted into database/performance/report.md.
3 — Confirm the Dialect	Write that your primary tools are PostgreSQL EXPLAIN / ANALYZE — not Oracle packages like DBMS_XPLAN.
4 — Mark Your Pass	Pass this exercise once every SQL TODO from Exercise 3 is filled in.

The hypothesis to write down

```
customer_id equality uses an index; LOWER(full_name) may seq scan.
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 6 before continuing: exercises/exercise-06-lab38-readiness.md. This is the last pre-lab check — once it passes, open lab38/LAB-38-GUIDE.md.

LAB OVERVIEW — SQL AND QUERY PERFORMANCE

Lab 38: PostgreSQL SQL and CRM Query Performance — DBMS_XPLAN, Indexes, Sargable Predicates

BUSINESS SCENARIO

- Before Spring Data JPA wires the CRM API to PostgreSQL (Lab 39), leadership freezes a performance gate: no 'add index because it felt slow' without actual plans, before/after numbers, and a report naming which query was measured.
- You own that gate for list-by-status, email lookup, customer→account joins, and deep paging for Amina (CUS-1001 ACTIVE), Ravi (CUS-1002 PROSPECT), and bulk synthetic rows.
- Guesswork indexes waste storage and slow writes — this lab makes you earn every retained index with evidence.

LEARNING OBJECTIVES

- Generate $\geq 50,000$ CRM rows and gather optimizer statistics.
- Capture actual plans with `DBMS_XPLAN.DISPLAY_CURSOR (ALLSTATS LAST)`.
- Create indexes for measured queries — not speculative ones.
- Rewrite non-sargable `TRUNC()` predicates into half-open date ranges.
- Compare nested-loop vs. hash-join strategies by selectivity.
- Implement deterministic `OFFSET/FETCH` and efficient keyset paging.
- Challenge every index: drop, re-plan, keep only justified ones.

WHAT YOU BUILD

- lab38-crm, copied from Lab 37: $\geq 50k$ customers with $\sim 70/30$ ACTIVE/PROSPECT skew; `DBMS_STATS` baseline; `ux_customer_email_norm` unique index; `ix_customer_status_created` composite index; a sargable date-range rewrite; `ix_account_customer` join index; deterministic `OFFSET + keyset` paging; an index challenge cycle; and `database/performance/report.md`.

KEY TAKEAWAY Fixtures CUS-1001 (Amina Khan, ACTIVE) and CUS-1002 (Ravi Singh, PROSPECT) anchor every measured query. Every retained index needs a before/after plan, buffer count, and median time.

LAB TASKS AND DELIVERABLES

PostgreSQL SQL and CRM Query Performance — implementation steps, deliverables, and the evaluation rubric.

#	Implementation Step
1	Branch Lab 37 → scaffold lab38-crm scripts 01–05 + report.md headings.
2	Generate ≥50k customers, ~70/30 ACTIVE/PROSPECT skew; preserve CUS-1001/CUS-1002.
3	Gather optimizer statistics; record NUM_ROWS / LAST_ANALYZED and query binds.
4	Capture the actual baseline email-lookup plan (DBMS_XPLAN, ALLSTATS LAST).
5	Create ux_customer_email_norm unique index; re-measure the identical query.
6	Create ix_customer_status_created composite index; measure the ACTIVE list query.
7	Rewrite TRUNC(created_at) into a sargable half-open range; compare plans.
8	Compare nested-loop vs. hash-join on customer→account; add ix_account_customer.
9	Implement OFFSET + keyset paging; challenge each index and write report.md.

Sargable date rewrite (Step 7)

```
-- Non-sargable: WHERE TRUNC(created_at) = DATE '2026-07-01';
-- Sargable:      WHERE created_at >= TIMESTAMP '2026-07-01 00:00:00'
--               AND created_at <  TIMESTAMP '2026-07-02 00:00:00';
```

EXPECTED DELIVERABLES

- database/performance/01–05 SQL scripts.
- ≥50k load with documented skew + preserved CRM fixtures.
- Baseline and after-index DBMS_XPLAN evidence.
- Sargable date rewrite comparison.
- Join strategy notes (selective vs. broad).
- Deterministic OFFSET + keyset paging demos.
- Index challenge cycle with retained-index justification.
- report.md with plan hash, buffers, median time, write cost.

RUBRIC (100 MARKS)

- Environment/Structure 10 · Core Implementation 30 · Integration/Config 15 · Failure Handling 15 · Automated Verification 10 · Security/Prod Awareness 10 · Documentation 10

KEY TAKEAWAY Adding indexes without before/after plans loses evidence marks; keeping every experimental index without a challenge loses production-awareness marks; forcing a 'pretty' plan without explaining why loses honesty marks.

MODULE 38 SUMMARY

In this module, we explored how to analyze, tune, and monitor SQL queries to ensure optimal database performance and a great user experience.

KEY CONCEPTS COVERED



SQL & Joins

SELECT, WHERE, ORDER BY, and INNER/LEFT/RIGHT/FULL OUTER JOIN.



Indexes & Plans

Clustered/non-clustered indexes and reading EXPLAIN ANALYZE plans.



Identifying Slow Queries

pg_stat_statements, detect-capture-analyze-verify process.



Tuning & Sargability

Query tuning strategies and avoiding non-sargable full table scans.



Efficient Design

Filter, join, select, and aggregate with the least resource usage.



Best Practices & Monitoring

Twelve production practices and continuous SQL monitoring.

THE PERFORMANCE CYCLE



KEY TAKEAWAY High-performing databases are built through smart design, efficient queries, proper indexing, and proactive monitoring. Apply these concepts consistently to deliver fast, reliable, scalable solutions.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of execution plans, performance metrics, and the most common causes and fixes for slow queries.

1. Which command is used in PostgreSQL to view the execution plan of a query?

- A. SHOW PLAN
- B. EXPLAIN**
- C. ANALYZE QUERY
- D. TRACE PLAN

3. What is a common cause of slow queries?

- A. Using too many indexes
- B. Missing or improper indexes**
- C. Small result sets
- D. Having constraints

2. Which metric indicates the total time a query took to complete?

- A. Total Cost
- B. Execution Time**
- C. Cache Hit Ratio
- D. Rows Scanned

4. Which of the following helps avoid full table scans?

- A. Using SELECT *
- B. Creating appropriate indexes**
- C. Storing data in text format
- D. Disabling statistics

KEY TAKEAWAY EXPLAIN shows the execution plan; execution time measures total query duration; missing or improper indexes are a leading cause of slow queries; appropriate indexes help avoid full table scans.

KNOWLEDGE CHECK (2 OF 2)

*Four more questions on SELECT *, join efficiency, why statistics matter, and the benefit of monitoring pg_stat_statements.*

5. Why is SELECT * considered a bad practice?

- A. It improves performance
- B. It retrieves unnecessary columns**
- C. It creates table locks
- D. It prevents index usage

7. Statistics on tables and indexes are important because they:

- A. Reduce storage size
- B. Help the optimizer choose better plans**
- C. Increase CPU speed
- D. Encrypt the data

6. Which join type is generally the most efficient when a valid relationship is required on both sides?

- A. CROSS JOIN
- B. LEFT OUTER JOIN
- C. INNER JOIN**
- D. FULL OUTER JOIN

8. What is the main benefit of monitoring pg_stat_statements?

- A. It modifies query results
- B. It provides insights into query performance**
- C. It creates indexes automatically
- D. It encrypts queries

KEY TAKEAWAY SELECT * retrieves unnecessary columns; INNER JOIN is typically the most efficient join for a required relationship; statistics help the optimizer choose better plans; pg_stat_statements provides insight into query performance.

MODULE 38 COMPLETE

SQL and Query Performance

You can now write efficient SQL, master all four JOIN types, use indexes wisely, read execution plans, and apply tuning strategies to keep PostgreSQL databases fast, reliable, and scalable.